Confidence Intervals and Margin of Error

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1. What is a confidence interval?

A confidence interval (CI) is a range of values used to estimate an unknown population parameter, such as the average height or the percentage of customers who like a product. We use a sample to estimate this range.

General form: Confidence interval = Sample estimate ± Margin of error

A 95% confidence level means that, if we repeated the sampling procedure many times and constructed a confidence interval each time, about 95% of those intervals would contain the true population value. It does not mean there is a 95% probability that the fixed true value lies inside this one calculated interval.

2. What is the margin of error?

The margin of error (MOE) tells us how far above and below our sample estimate the confidence interval extends. A smaller margin of error gives a narrower, more precise interval, assuming the method is appropriate.

For a population mean with known population standard deviation: $MOE = z^* \times (\sigma / \sqrt{n})$

Where,

- z^* is the critical value (1.96 for a 95% confidence level)
- σ is the population standard deviation
- n is the sample size.

The quantity $\sigma / \sqrt{n}$ is the standard error.

When the population standard deviation is unknown, we typically use the sample standard deviation and a t critical value instead, particularly for small samples.

3. Example 1: Average test score

Suppose a random sample of 100 students has an average score of 70 marks. The population standard deviation is known to be 10 marks. Construct an approximate 95% confidence interval for the population mean.

Step 1: Sample mean = 70; $\sigma = 10$; $n = 100$; $z^* = 1.96$.

Step 2: Standard error = $10 / \sqrt{100} = 1$ mark.

Step 3: Margin of error = $1.96 \times 1 = 1.96$ marks.

Step 4: Confidence interval = $70 \pm 1.96 = (68.04, 71.96)$ marks.

Interpretation: Using this method, we report a 95% confidence interval from 68.04 to 71.96 marks for the true average score.

4. Example 2: Percentage of satisfied customers

In a random survey of 400 customers, 240 say they are satisfied. The sample proportion is 240 / 400 = 0.60, or 60%. We want an approximate 95% confidence interval for the true satisfaction percentage.

Step 1: $\hat{p} = 0.60$; $n = 400$; $z^* = 1.96$.

Step 2: Standard error = $\sqrt{[0.60 \times 0.40 / 400]} \approx 0.0245$.

Step 3: Margin of error = $1.96 \times 0.0245 \approx 0.048$, or 4.8 percentage points.

Step 4: Confidence interval ≈ 60% ± 4.8% = (55.2%, 64.8%).

Interpretation: We estimate that about 55.2% to 64.8% of all customers are satisfied, using a 95% confidence interval.

5. What changes the margin of error?

Factor	**Typical effect**
Larger sample size (n)	Smaller margin of error; more precise estimate.
Higher confidence level (e.g., 99% vs. 95%)	Larger margin of error; wider interval.
More variability in the data	Larger margin of error; wider interval.

6. Quick comparison and key points

Term	**Meaning**
Sample estimate	Single best estimate from the observed sample (e.g., 60%).
Margin of error	Amount added to and subtracted from the estimate (e.g., ±4.8 points).
Confidence interval	The complete lower-to-upper range (e.g., 55.2%–64.8%).

Remember: Confidence intervals express sampling uncertainty, but they do not automatically correct biased samples, misleading questions, missing data, or other systematic errors.